

Expressions & Equations

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Order of operations: do the work inside **grouping symbols ()** first, then **multiply and divide**, then **add and subtract**. Example: $2 \times (3 + 4) = 2 \times 7 = 14$. A **letter or box** can stand for an unknown number. To find it, ask what value makes the equation true. Example: $5 + \square = 12$, so $\square = 7$.

Round 1 — Retrieve It

1. Which step do you do first: grouping symbols () or adding?

2. After grouping symbols, which comes next: multiply/divide or add/subtract?

3. What is $2 \times (3 + 4)$?

4. In $5 + \square = 12$, what does the box stand for?

Round 2 — Sketch It

Write out the steps for $2 \times (3 + 4)$. Circle the part you do first, then show the final answer.

Expressions & Equations — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. Solve $2 \times (3 + 4)$. Show the step inside the parentheses first.

2. Solve $10 - 2 \times 3$. Which operation do you do first, and why?

3. Find the unknown: $5 + \square = 12$. What is $\square$? Tell how you found it.

Round 4 — Reason from Evidence

A student solved $2 \times (3 + 4)$ by doing 2×3 first and got 10. Explain the order-of-operations mistake and give the correct answer.

Expressions & Equations — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate

Explain the order of operations in your own words. Use the words grouping, multiply, and add.

C

Connect

Connect a box or letter for an unknown to a real question, like "how many more do I need?" Write an equation for it.

E

Extend

Solve $3 \times (8 - 2)$ and find the unknown in $4 \times \square = 20$. Show your steps for both.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Expressions & Equations

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Grouping symbols () first.
2. Multiply/divide (before add/subtract).
3. 14 ($3 + 4 = 7$, then $2 \times 7 = 14$).
4. The unknown (missing) number.

Sketch It — look for

Steps shown as $2 \times (3 + 4) = 2 \times 7 = 14$, with the $(3 + 4)$ circled as the first step.

Apply It

1. $2 \times (3 + 4) = 2 \times 7 = 14$.
2. $10 - 2 \times 3 = 10 - 6 = 4$ (multiply 2×3 first, then subtract).
3. $\square = 7$ (because $12 - 5 = 7$; $5 + 7 = 12$).

Reason from Evidence — look for

Grouping symbols come first, so you must add $3 + 4 = 7$ before multiplying. The correct answer is $2 \times 7 = 14$, not 10.

ACE — Extend look-for

$3 \times (8 - 2) = 3 \times 6 = 18$; and $4 \times \square = 20$ gives $\square = 5$ (because $20 \div 4 = 5$).

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.